

Heat Transfer

Final Lab Report
Unit Operations Lab 1
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Group 1
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1. Executive Summary (Examples at end of the Guideline) (10 pts)

The objective of this experiment was to quantify the heat transfer performance of a steam jacketed, agitated tank to obtain design-relevant heat transfer coefficients for a well-mixed vessel. Water was heated by condensing steam in the jacket, and heat was removed with an internal coil and an external recirculation loop with a heat exchanger. Overall heat transfer coefficients based on the inner wall area () and the liquid-side convective heat transfer coefficient () for water were determined by operating the system at steady state, measuring temperatures at the tank, coil, heat exchanger, and condenser/steam trap, recording relevant flow rates, and using condensate collection to determine the steam heat input. The main parameters evaluated were impeller speed and baffle configuration (baffled versus unbaffled), since both directly affect mixing intensity and circulation near the heat transfer surface, and therefore the liquid-side film resistance.

In the unbaffled cases, the reported steady-state coefficients decreased with increasing speed across the three runs: at 340 rpm, U was approximately $1.13 \times 10^4 \text{ W/m}^2 \cdot \text{K}$ and h_i was approximately $1.18 \times 10^4 \text{ W/m}^2 \cdot \text{K}$; at 520 rpm, U was approximately $6.87 \times 10^3 \text{ W/m}^2 \cdot \text{K}$ and h_i was approximately $7.03 \times 10^3 \text{ W/m}^2 \cdot \text{K}$; and at 745 rpm, U was approximately $2.63 \times 10^3 \text{ W/m}^2 \cdot \text{K}$ and h_i was approximately $2.65 \times 10^3 \text{ W/m}^2 \cdot \text{K}$. In the baffled cases, the reported coefficients were much smaller in magnitude and showed a modest increase at the highest speed: U and h_i were about 59–60 $\text{W/m}^2 \cdot \text{K}$ at 340–500 rpm, increasing to about 76–77 $\text{W/m}^2 \cdot \text{K}$ at 694 rpm.

Across the recirculation-flow trials, U and h_i varied with flow rate and configuration but showed noticeable scatter rather than a single smooth trend. A major limitation affecting consistency was restricted heating from clogged building steam, which reduced the tank's ability to reach and hold the target temperature; to compensate, cooling water was reduced and at times shut off, so calculations treated cooling-water heat removal as minimal, increasing uncertainty in the energy balance and contributing to variability in the computed coefficients.

2. Introduction (10 pts)

Continuous stirred-tank reactors (CSTRs) and other agitated, jacketed vessels are widely used in industry because strong mixing produces nearly uniform composition and temperature in the bulk liquid, which improves product consistency and helps prevent hot spots, phase changes, or undesired side reactions. In these systems, temperature is commonly controlled by adding heat through a steam jacket and removing heat using internal coils or an external recirculation loop with a heat exchanger. Because heat removal or addition can limit both productivity and safety, reliable heat transfer performance is essential for the design, scale-up, and operation of stirred tank equipment.

The objective of this experiment was to determine the heat transfer performance of a steam jacketed agitated tank by calculating the overall heat transfer coefficient based on the inner wall area, U , and the tank side convective heat transfer coefficient, h_i , for water.

The experiment also examined how key operating and mechanical parameters influence heat transfer, specifically impeller speed, baffle configuration, and operating temperature. These parameters are directly tied to mixing intensity and flow patterns near heat-transfer surfaces, which strongly affect the convective film resistance on the liquid side.

The apparatus consists of a stirred vessel equipped with an agitator, a steam jacket for heating, and an internal cooling coil for heat removal. The system also included an external recirculation loop in which tank liquid was pumped through an external heat exchanger and returned to the vessel. Removable baffles allowed comparison of behavior between baffled and unbaffled operations. Temperatures were monitored at the tank, the coil inlets and outlets, the external heat exchanger, and the steam condenser/steam trap system used to quantify the steam condensation rate. Rotameters measured water flow through the coil and heat exchanger, and a tachometer measured the impeller speed.

During operation, the tank was brought to steady state and held within a moderate temperature range while variables were adjusted one at a time. Steady state was identified by a stable tank temperature over several minutes, after which the required flow rates, temperatures, and steam condensate collection data were recorded. Trials were conducted for multiple impeller speeds and for both baffled and unbaffled configurations to establish trends in and with mixing conditions. The results of this experiment can be related to relevant industrial stirred tank applications, such as solution blending and continuous reaction systems, where the heat transfer capacity and temperature control can strongly influence product quality and safe operation.

3. Theory (10 pts)

The purpose of the experiments is to examine heat transfer from the condensing steam in the steam jacket to the scale wall, heat transfer across the solid wall, and heat transfer from the wall-scale of the stirred tank to the vessel fluid, which in this case is water [6].

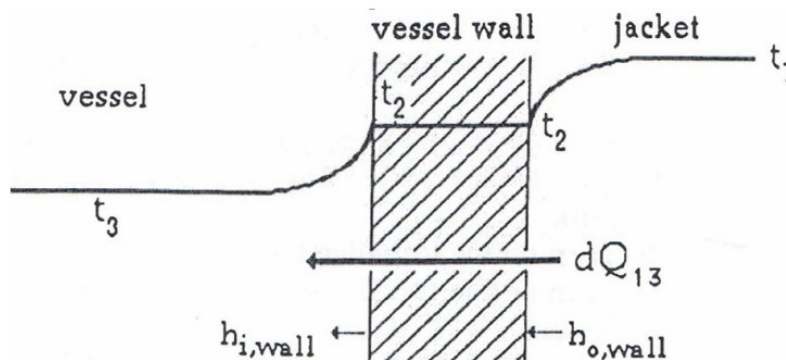


Figure 1: Heat transfer across a vessel wall

The governing equation for heat transfer models is expressed by Fourier's Law

In the experiment, the following assumptions are made when calculating the heat transfer coefficients: no scale on either side of the solid wall, negligible resistance to heat transfer through the wall, and a wall thickness small compared to the tank diameter. Because there is negligible resistance, heat transfer by the wall, and the wall thickness is small, the temperature across the x direction is nearly zero, making equation 1 the following

Therefore, Newton's law of cooling is applied instead, and is described in equation 3

Where the change in departure is dependent on time and equals the thermal conductivity times the change in temperature of the ambient air minus the surface temperature.

Another assumption for this experiment is steady state, as the calculations use data after the system has reached steady state; therefore, equation 3 becomes equation 4.

The equation describes a single fluid exchanging heat with the surroundings at steady state. However, in this experiment, heat transfer occurs between two fluids: steam and water. Therefore, equation 5 is used instead

Equation 5 describes heat transfer between two fluids separated by a wall; again, assuming steady state, it simplifies to equation 6.

Equation 6 can be rewritten as the following

Where the right-hand side of the equation is the heat transfer from T1 to T2, and the left-hand side equation is the heat transfer from T2 to T3. Therefore, the heat transfer from T1 to T3 can be written in the following way, as in equation 7, where the following expression is developed

Where h_i and h_o are the heat transfer coefficients of the inner and outer walls. A_i is the inner surface area and A_o is the outer surface area. Equation 8 essentially states that, at steady state, the heat flux is the same through each stage of wall resistance.

Because the wall thickness is small compared to the tank diameter, the inner and outer wall areas are roughly equal, and the heat transfer coefficients can be expressed as follows.

So that

Equation 10 allows us to calculate using either of two methods, the first one being using a steam energy balance (shown in equation 11) or a stirred tank fluid energy balance (shown in equation 12).

Once is calculated, all other parameters can be calculated in this experiment.

Empirical correlations allow for comparison of the measured coefficients with literature values. For an unbaffled stirred tank containing a Newtonian fluid, the following empirical correlations exist.

And

Where is the tank diameter, is the impeller diameter, is the viscosity of the fluid, is the density of the fluid, k is the thermal conductivity of the fluid, is the specific heat capacity of the fluid, and n is the speed or rotation (rpm) of the impeller.

In order to calculate the heat transfer coefficient of the outer wall is by using equation 14

Where valid when

4. Results (15 pts)

340 RPM						
Time,	RP	Q13 J/s	U	h0,	hi J/(s*m ²	hi from nut,

min	M		$(J/s \cdot m^2 \cdot K)$	$J/(s \cdot m^2 \cdot K)$	K	$(J/(s \cdot m^2 \cdot K))$
0	310	134871.7	6603.6	300522.1	6751.9	3202.5
5	317	163369.6	8541.2	300522.1	8791.0	3288.7
10	323	164874.1	8922.3	300522.1	9195.3	3366.1
15	330	172845.8	10253.1	300522.1	10615.2	3578.6
SS	340	176396.4	11335.7	300522.1	11780.1	3762.0

Table 1: Calculations for no baffles at 340 RPM

520 RPM						
Time, min	RP M	Q13 J/s	U $(J/s \cdot m^2 \cdot K)$	h0, $J/(s \cdot m^2 \cdot K)$	hi $J/(s \cdot m^2 \cdot K)$	hi from nut $J/(s \cdot m^2 \cdot K)$
0	489	104827.0	6466.9	315375.8	6602.4	4698.6
5	500	108041.9	6408.9	315375.8	6541.9	4720.8
10	513	115746.6	6155.7	315375.8	6278.3	4629.9
15	517	122451.7	6626.6	315375.8	6768.8	4678.8
SS	520	124784.8	6873.4	315375.8	7026.6	4722.1

Table 2: Calculations for no baffles at 520 RPM

745 RPM						
Time, min	RPM	Q13 J/s	U $(J/s \cdot m^2 \cdot K)$	h0, $J/(s \cdot m^2 \cdot K)$	hi $J/(s \cdot m^2 \cdot K)$	hi from nut $J/(s \cdot m^2 \cdot K)$
0	720	33500.06	1812.88	326246.86	1823.01	5846.93
5	731	40606.09	2122.94	326246.86	2136.85	5818.89
10	738	47680.26	2451.24	326246.86	2469.80	5806.18
15	741	52131.15	2593.61	326246.86	2614.39	5757.39
SS	745	52859.89	2629.87	326246.86	2651.24	5778.09

Table 3: Calculations for no baffles at 745 RPM

340 RPM						
Time, min	RPM	Q13 J/s	U $(J/s \cdot m^2 \cdot K)$	hi $J/(s \cdot m^2 \cdot K)$	h0 $J/(s \cdot m^2 \cdot K)$	
0	337	1502.5	65.3	65.4		35622.2
5	335	1542.0	68.3	68.5		35622.2

10	338	1647.5	73.6	73.8	35622.2
15	338.7	1502.5	67.9	67.9	35622.2
SS	340	1317.9	59.8	59.9	35622.2

Table 4: Calculations for baffles at 340 RPM

500 RPM					
Time, min	RPM	Q13 J/s	U (J/s*m ² *K)	hi J/(s*m ² K)	h0 J/(s*m ² K)
0	489	1449.8	61.3	61.4	36388.2
5	493	1634.3	69.3	69.4	36388.2
10	497.9	1410.2	61.3	61.3	36388.2
15	500	1449.8	64.2	64.2	36388.2
SS	500	1317.9	58.9	58.9	36388.2

Table 5: Calculations for baffles at 500 RPM

694 RPM					
Time, min	RPM	Q13 J/s	U (J/s*m ² *K)	hi J/(s*m ² K)	h0 J/(s*m ² K)
0	690	1449.8	61.4	61.5	34913.3
5	691	1449.8	62.1	62.2	34913.3
10	691.3	1515.7	66.2	66.3	34913.3
15	693	1660.6	73.2	73.3	34913.3
SS	694	1713.3	76.5	76.6	34913.3

Table 6: Calculations for baffles at 694 RPM

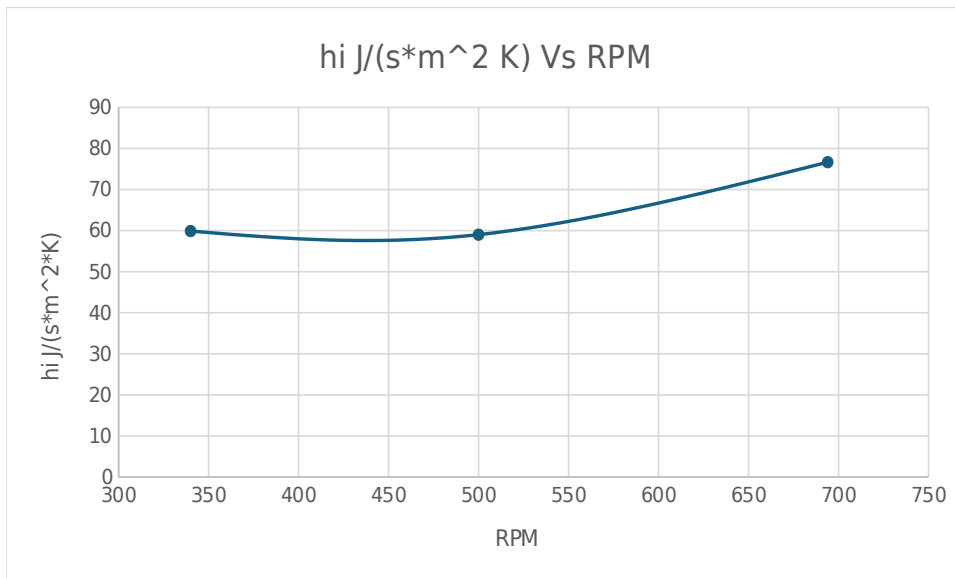


Figure 2: hi vs RPM with baffles

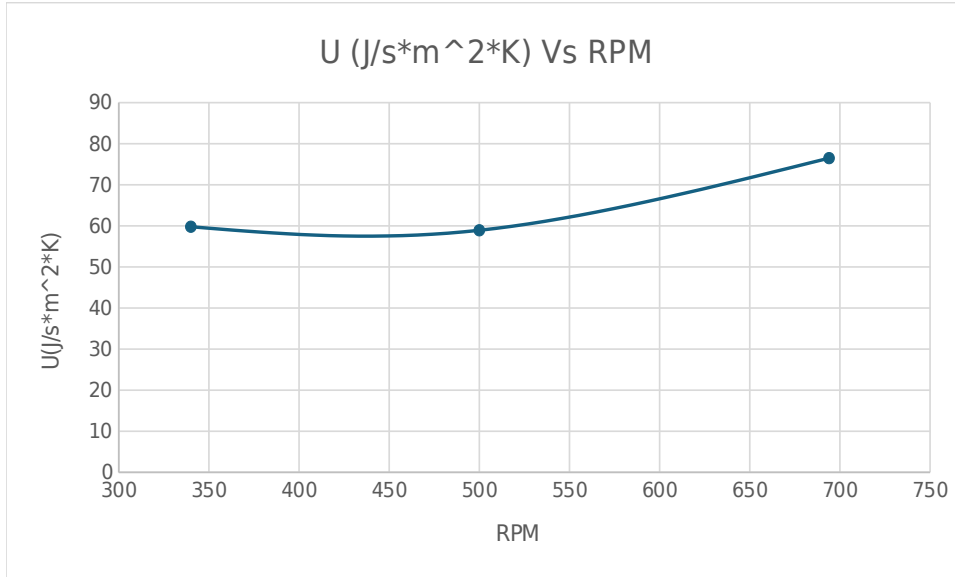


Figure 3: U vs RPM with baffles

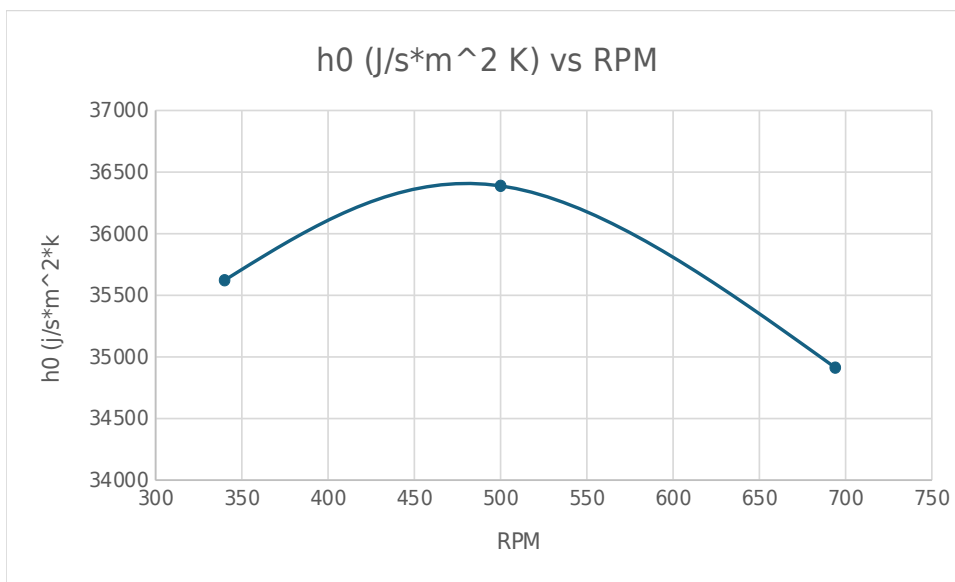


Figure 4: h_0 vs RPM with baffles

From Figures 2–4, clear trends emerge between impeller speed and the various heat transfer coefficients. Figure 2 shows that the inner-wall coefficient, U_i , increases with increasing impeller speed. This behavior is expected because higher impeller speed increases circulation, turbulence, and boundary-layer disruption in the tank, which strengthens convective heat transfer between the bulk liquid and the inner wall. Figure 3 shows that the overall coefficient based on inner area, U , increases with impeller speed in a manner similar to U_i . This follows directly from Eq. (9),

since is limited by the larger thermal resistance. At low rpm, is relatively small, so dominates and remains low. As rpm increases, rises and internal resistance decrease, causing to increase accordingly. At high rpm, the internal convection is strong enough that changes in have a smaller influence on compared to changes in .

Figure 4 indicates that the outer-wall coefficient, , increases from low rpm to an intermediate maximum, then decreases at the highest rpm. Since the impeller primarily affects the internal convection, a non-monotonic trend is often best interpreted as an indirect effect or a sensitivity of the calculation. At the highest rpm, internal heat transfer becomes very strong, so any small uncertainties in temperature measurements or assumptions used to back-calculate can appear as a reduction in the computed outer coefficient. Overall, the presence of baffles supports these trends by suppressing vortex formation and improving mixing, which increases turbulence and significantly enhances internal heat transfer, driving the observed increase in and with impeller speed.

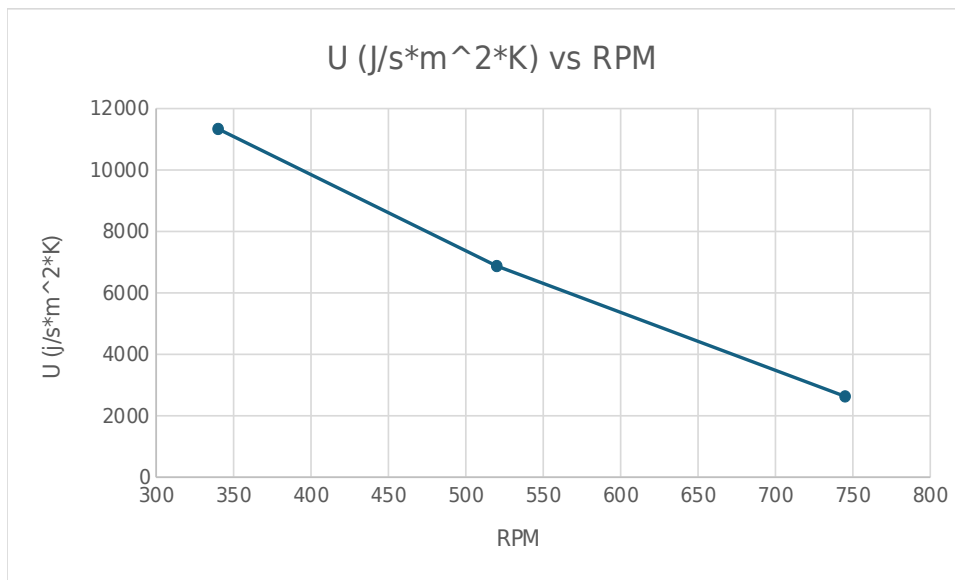


Figure 5: U_i vs RPM no baffles

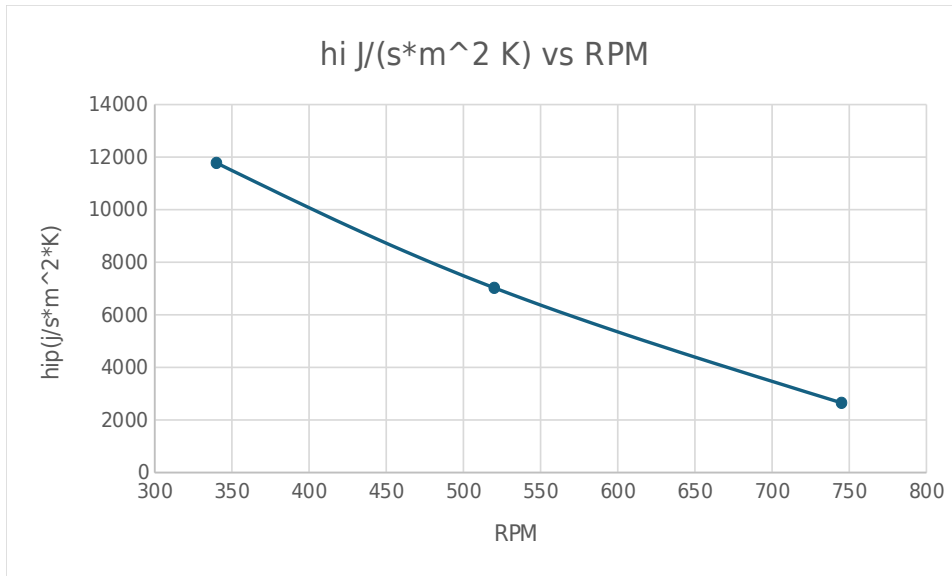


Figure 6 hi vs RPM no baffles

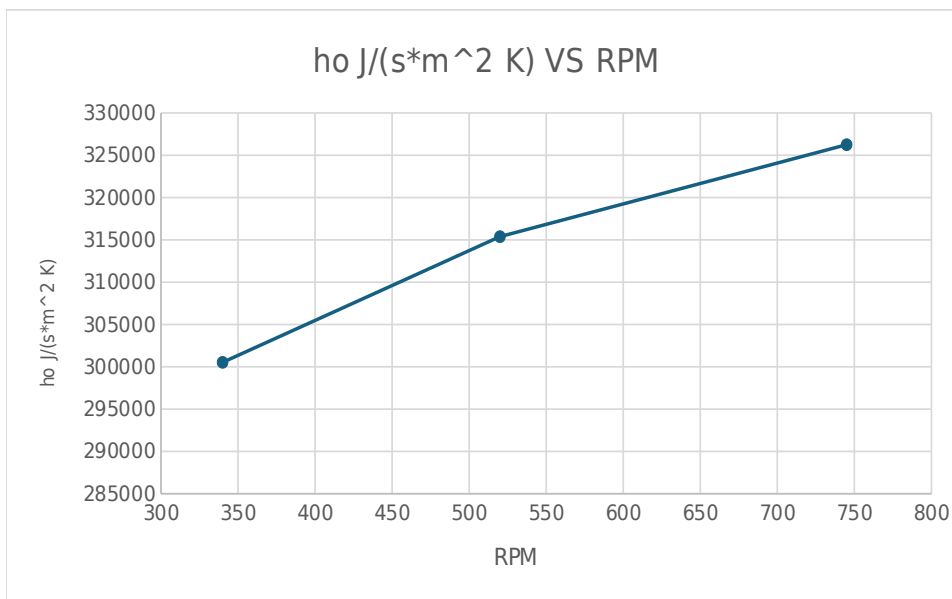


Figure 7: ho vs RPM no baffles

Figures 5–7 describe the heat transfer behavior at different impeller speeds when no baffles are used in the vessel. Figure 7 shows that the outer-wall heat transfer coefficient, h_o , increases with rpm. This trend is reasonable because increasing agitation can still raise the overall energy transport near the wall and can increase the apparent external-side performance when the steam-side conditions remain steady. Figure 6 shows the inner-wall heat transfer coefficient, h_i , decreasing as rpm increases. Without baffles, the impeller is more likely to form a swirling vortex and bulk rotation rather than strong top-to-bottom circulation. That type of flow can reduce effective mixing and weaken the renewal of fluid at the inner wall. As a result, the liquid can develop stronger temperature gradients (some regions hotter, others colder), which lower the effective driving force for convection at the inner wall and lead to a smaller calculated h_i . Figure 5 shows that the overall heat transfer coefficient, U_i , follows the same general trend as h_i . This is consistent with Eq. (9), because the overall coefficient is controlled by the larger resistance term. In this case, h_i remains smaller than h_o across the rpm range, so h_i dominates the denominator and changes in h_i have the greatest impact on U_i , even though h_o increases. Overall, the results indicate that removing baffles significantly reduces internal heat transfer performance, due to poorer mixing and weaker turbulence near the inner wall.

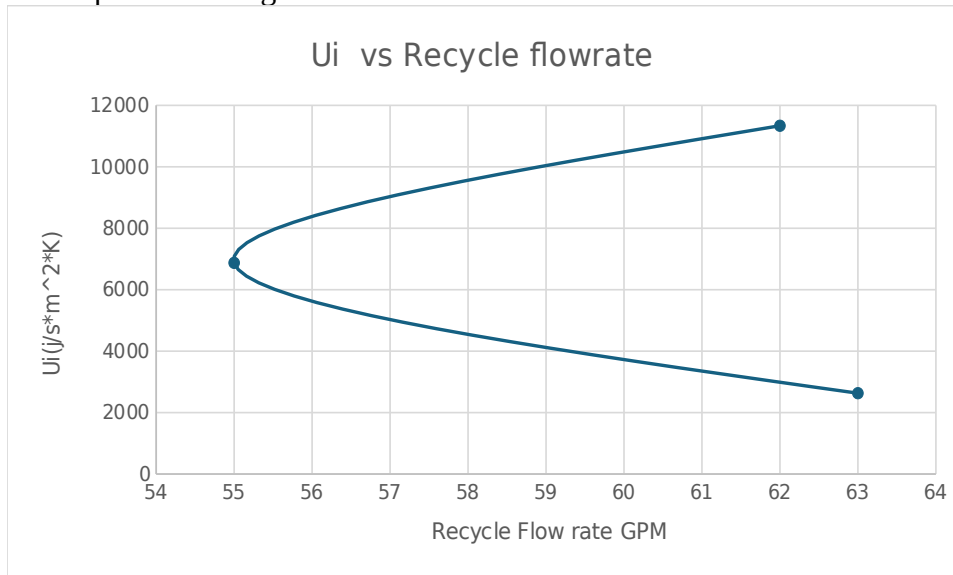


Figure 8: U_i vs recirculation flow no baffles

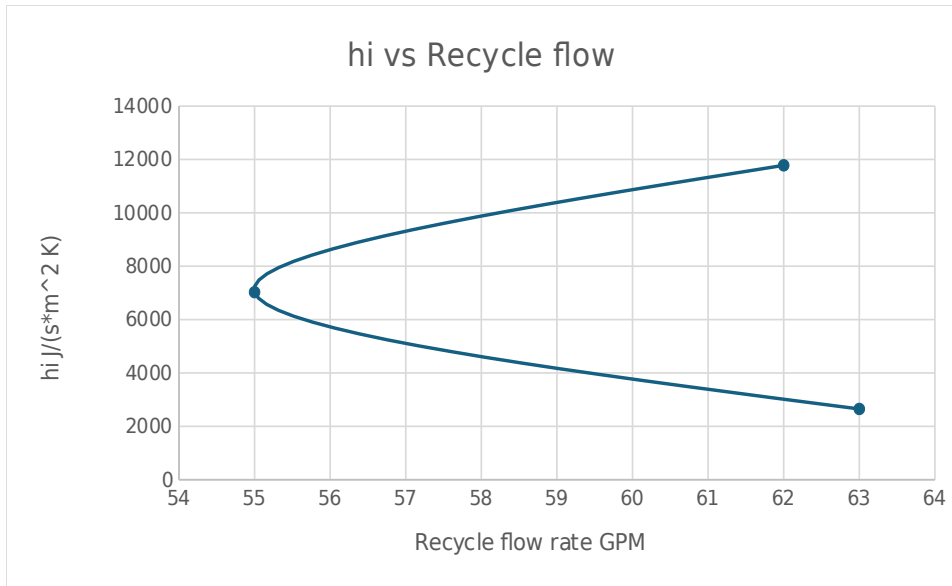


Figure 9: h_i vs recirculation flow rate no baffles

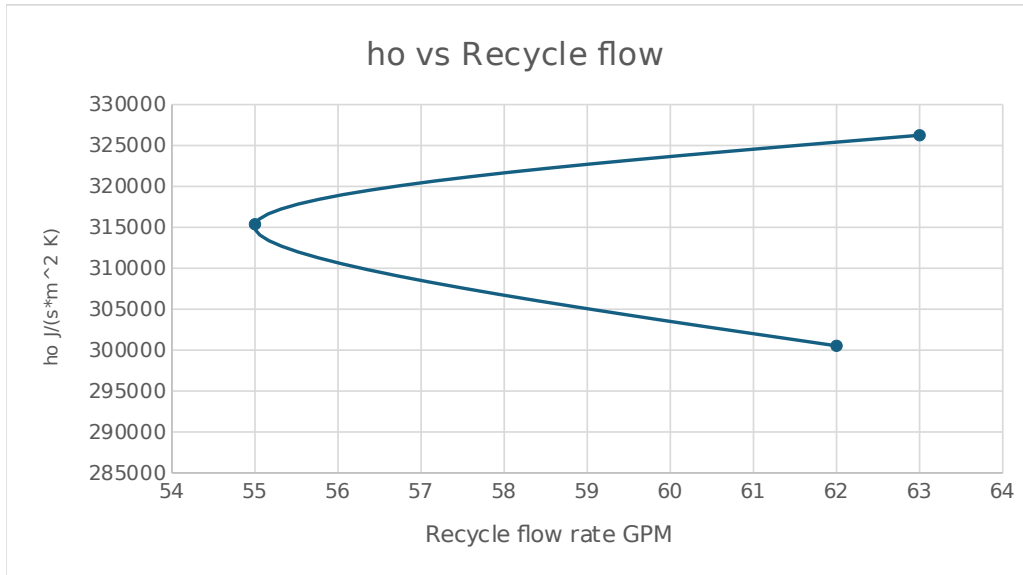


Figure 10: h_o vs Recirculation flow rate no baffles

Figures 8, 9, and 10 show the relationship between the heat transfer coefficient and the recirculation flow rate. All figures show the same relationship between the heat transfer coefficient and the recirculation flow rate. In an unbaffled vessel, increasing flow does not necessarily mean better heat transfer because higher flow can cause turbulence and make mixing less effective. As a result, the internal heat transfer coefficient is lower and still dominates; in this case, the overall heat transfer coefficient follows the same trend.

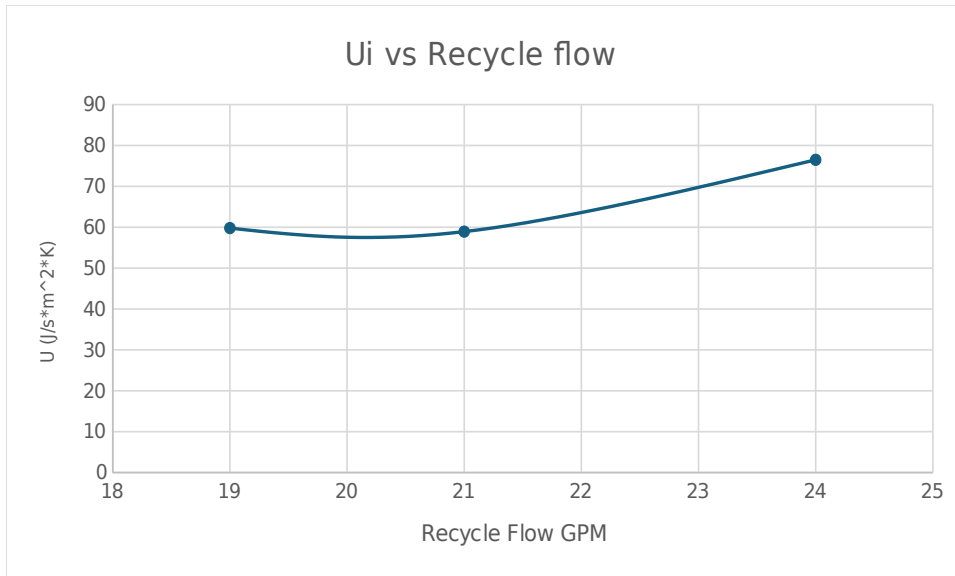


Figure 11: U_i vs Recirculation flow rate with baffles

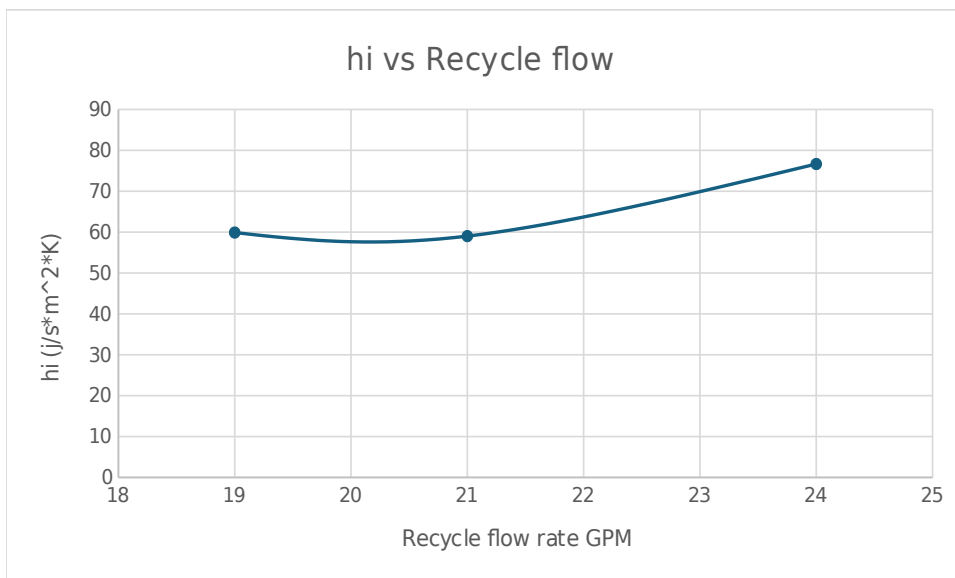


Figure 12: U_i Vs. Recirculation flow rate with baffles

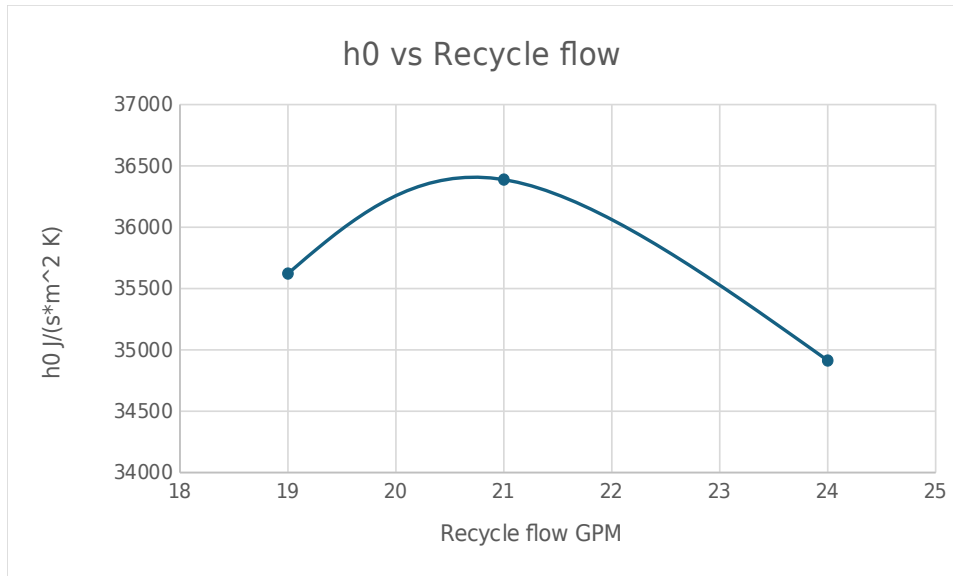


Figure 13: h_0 Vs Recirculation flow rate with baffles

Figures 11, 12, and 13 show the relationship between the heat transfer coefficients and the recirculation flow rates for the baffled system. In the baffled system, increasing the recycle flow rate increases the value, since the baffles manage the turbulent flow and allow for better mixing, allowing the increased recycle flow to enhance wall sweeping and increase internal convective heat transfer. For h_0 , it initially increases with increased flow due to better condensation, but it decreases at higher recycle flow rates, possibly because more condensate forms, meaning more vapor becomes liquid and lowers its heat transfer capability.

5. Discussion (15 pts)

To begin the discussion, a few things shall be noted in response to the results section. During the lab, a few variables had to be altered for the following reasons. The cooling water was at a very low temperature due to the cold weather. As a result, the tank vessel never reached the 65 °C–85 °C range. In response, the cooling water was shut off, and its temperature and flow were neglected. Secondly, the steam piping supplying the system was clogged with minerals and similar debris, resulting in a very low inlet steam flow. Thus, this didn't allow the system to fully be entrapped in the steam and caused difficulties in maintaining a temperature. Due to both constraints in the lab, the vessel's temperature kept dropping and wouldn't increase, making it difficult to establish a steady-state temperature in the tank. Lastly, the water loaded into the tank vessel was also very cold, causing a very cold system and making it hard to heat up and maintain a temperature.

As the RPM increased, the Reynolds' Number, Re , and Nusselt Number, Nu , increased. This is due to increasing the impeller speed from 340 RPM to 745 RPM; the speed

increase will improve mixing in the tank, affecting slurry formation and making the flow more aggressively turbulent. The Re is determined by the ratio of inertial to viscous forces in the liquid (water); it is relatively high, indicating that inertial forces dominate viscous forces [1]. As the Re increases, the liquid velocity increases, as evidenced by the impeller's RPM. Additionally, Nu describes the flow of water based on convective heat transfer, length, and thermal conductivity [1]. A high Nu was observed in the results, indicating strong near-wall mixing and strong convection. In turbulent flow, thin boundary layers enable very efficient heat transfer. So, the results for Re and Nu match the literature because the trends are consistent with it.

In the experiment, the Prandtl Number, Pr , describes the ratio of momentum diffusivity to thermal diffusivity [2]. Pr also says that thermal diffusion occurs slightly slower than momentum diffusion [2]. This is affected by the relative thickness of the wall boundaries and the limitations of possible heat transfer. In the results, a steady Pr value is observed, as water has a very small range of 5-7 Pr [2]. Pr is often treated as constant in water over a temperature range of 20 °C-60 °C [2]. In this lab, the temperature conditions fall within that range, correlating with the steady Pr calculated.

As for the , with the baffles is 0 due to the cooling water being turned off, causing . Due to this, when calculating the values didn't change much after losing the cooling-water effect. Steady values in each RPM trial are shown, but as the RPM increases, the increases. This is due to greater mixing, which increases heat transfer in the tank vessel. Additionally, as the RPM increases, the values of increase and decrease, respectively. Since, is defined as equation 9, the correlation between can be shown as a linear correlation. In which the overall heat transfer becomes more efficient due to the heat being transferred to more efficiently, which is allowing to be correlated with as shown in Figure 1.

As for the , with no baffles isn't 0 due to the cooling water being used at a low flow rate, causing . In turn, comparing the results between baffles and no baffles, the values showed some variance when compared to steady values with the baffles. This can be due to the addition of affecting overall heat transfer due to extremely cold cooling water and a clogged steam supply, which inefficiently mixes hot and very cold feed, causing the temperature to constantly drop and plateau. By changing the impeller speed, the values of decreased due to the impeller speed increasing the mixing of the cooling water in the system, creating a worse heat transfer as the RPM increased. Additionally, rather than decreasing it increased with showing that the system wasn't diffusing heat efficiently and allowing to steadily increase instead of being correlated with described in Figure 1.

The results didn't yield issues; instead, issues were yielded from uncontrollable variables such as weather. This gave a new perspective on the heat transfer lab by neglecting the cooling water, allowing the system to be regulated by steam flow and recirculation flow. This created a temperature issue, as the water in the vessel was also very cold, making it very hard to maintain a consistent temperature between baffled and non-baffled sections.

All in all, the changes made to this system show its effects and the influence of its surroundings, and how processes can be affected by ambient conditions. This shows the importance of plant location and how process engineers may struggle in different environments where heat transfer is prevalent. This raises the question of how warmer temperatures may affect the system later in the semester, and how the heat transfer would be affected. [3]

6. References (10 pts)

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7. Appendix I: Data Tabulation/Graphs (10 pts)

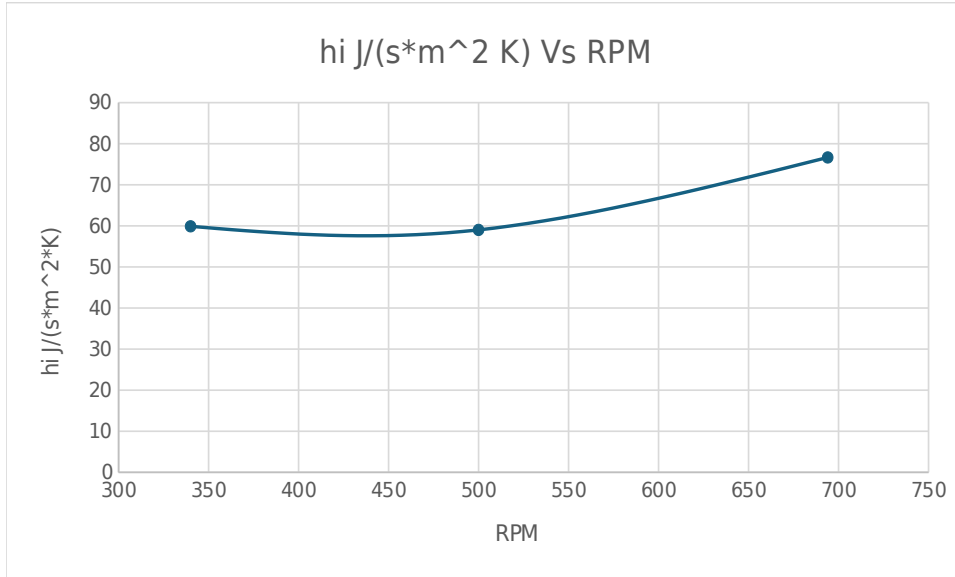


Figure 14: hi vs RPM with baffles

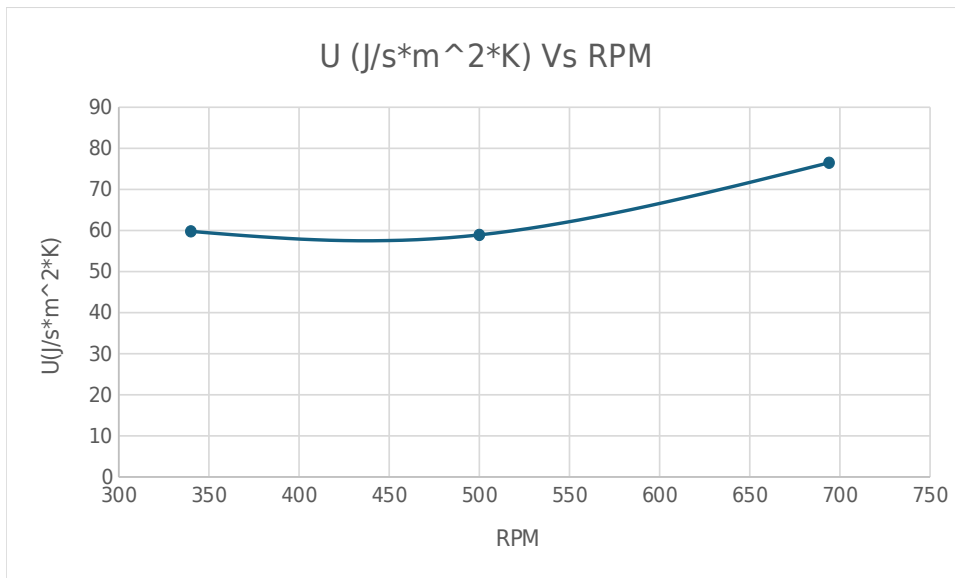


Figure 15: U vs RPM with baffles

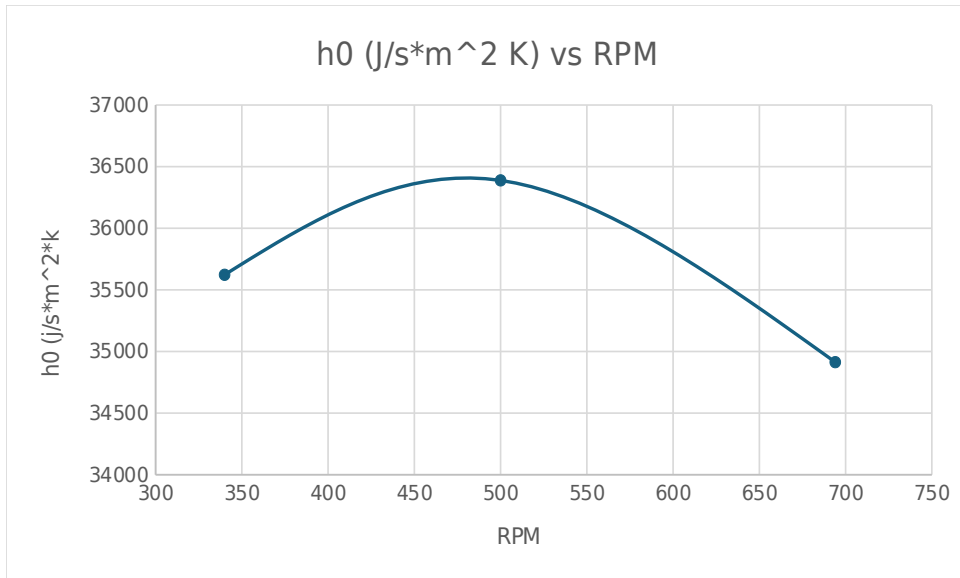


Figure 16: h_0 vs RPM with baffles

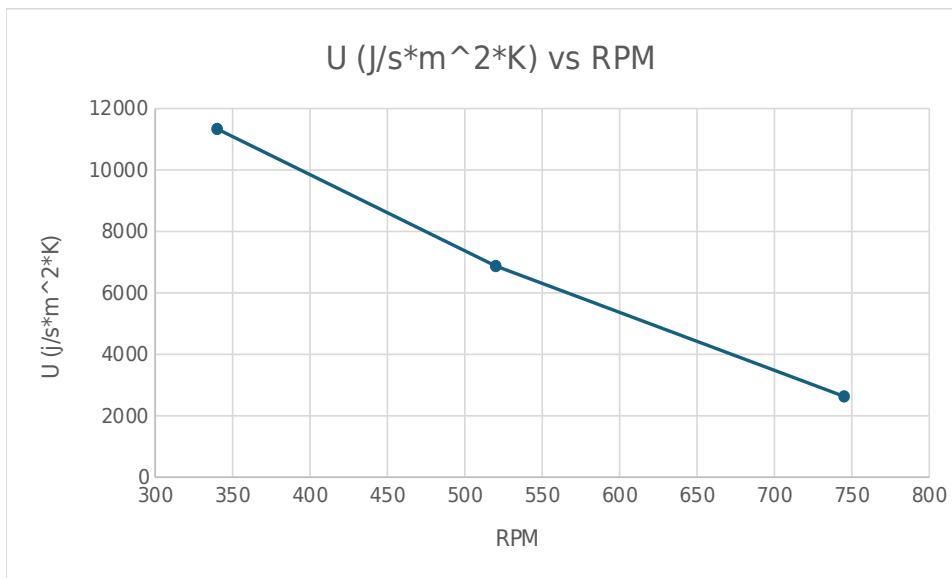


Figure 17: U_i vs RPM no baffles

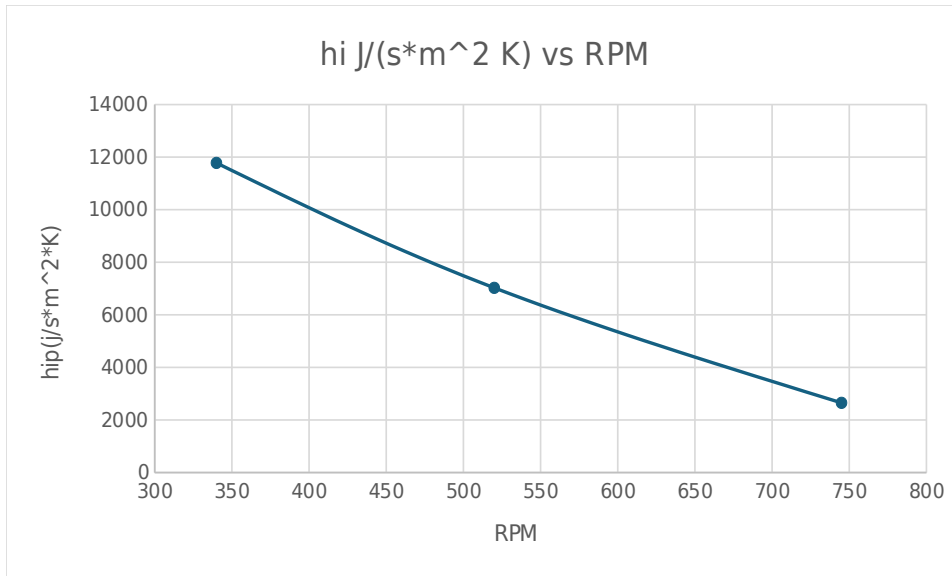


Figure 18 hi vs RPM no baffles

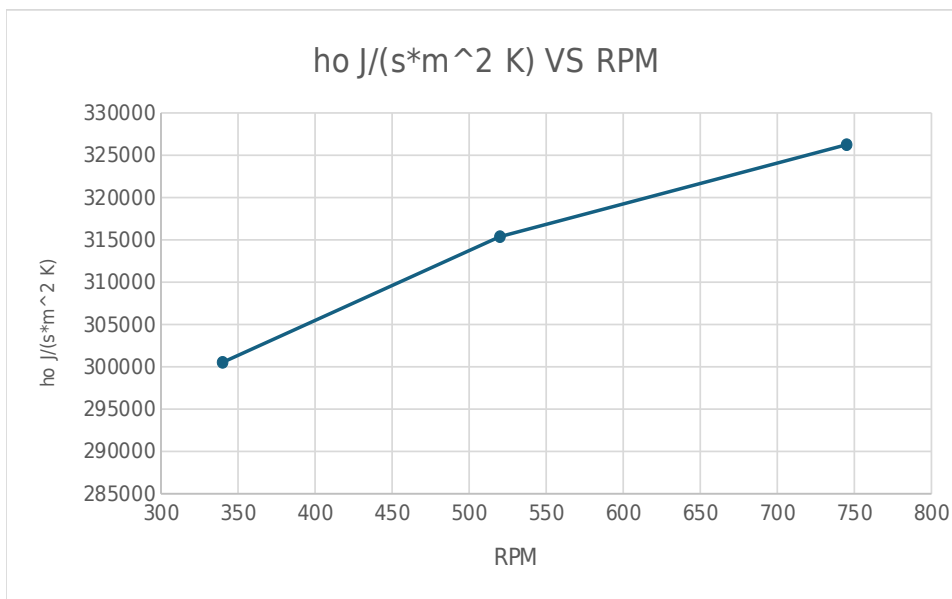


Figure 19: ho vs RPM no baffles

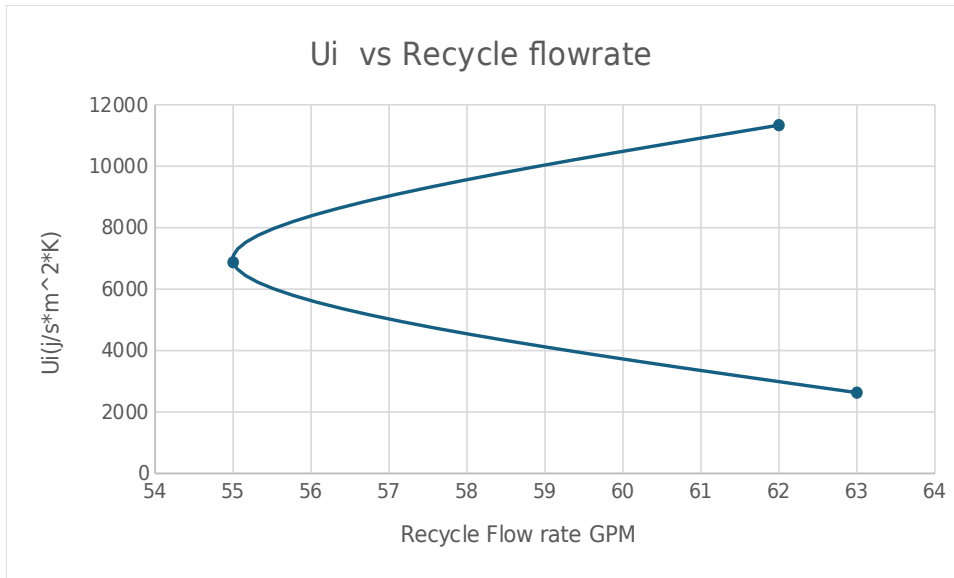


Figure 20: U_i vs recirculation flow no baffles

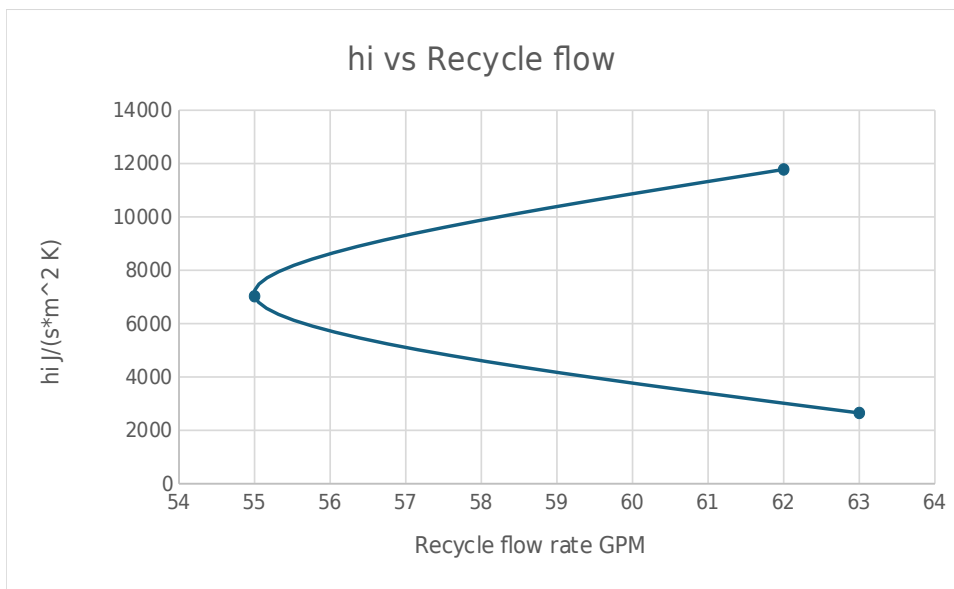


Figure 21: h_i vs recirculation flow rate no baffles

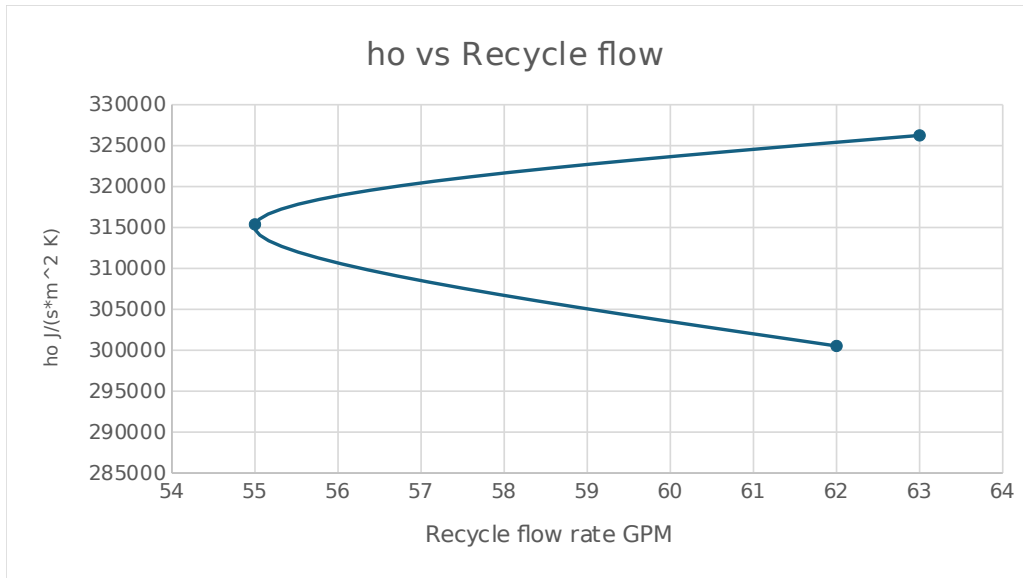


Figure 22: ho vs Recirculation flow rate no baffles

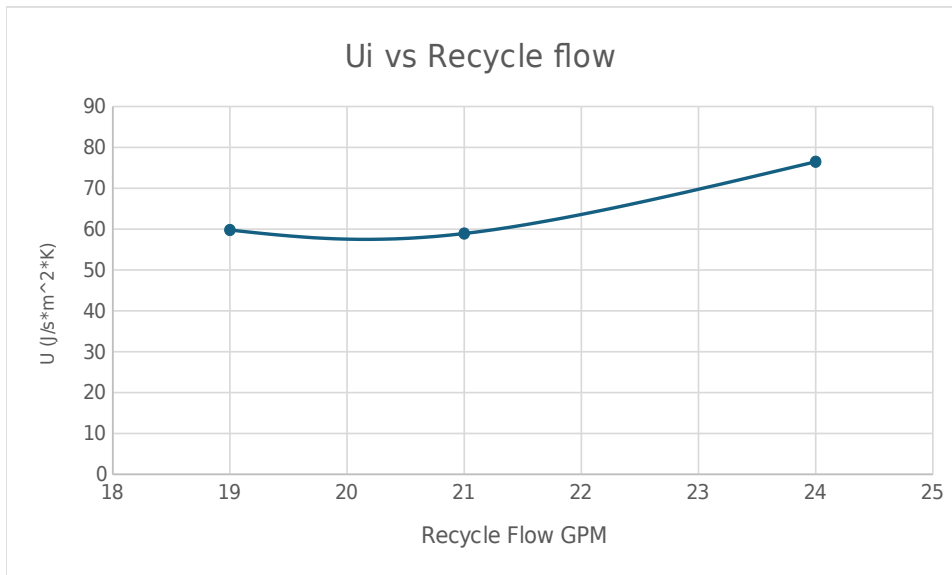


Figure 23: Ui vs Recirculation flow rate with baffles

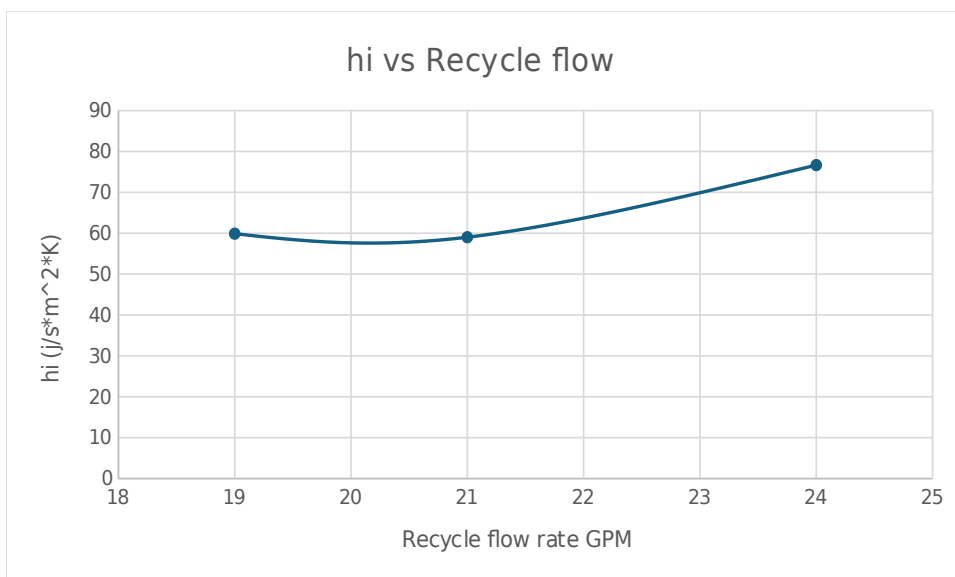
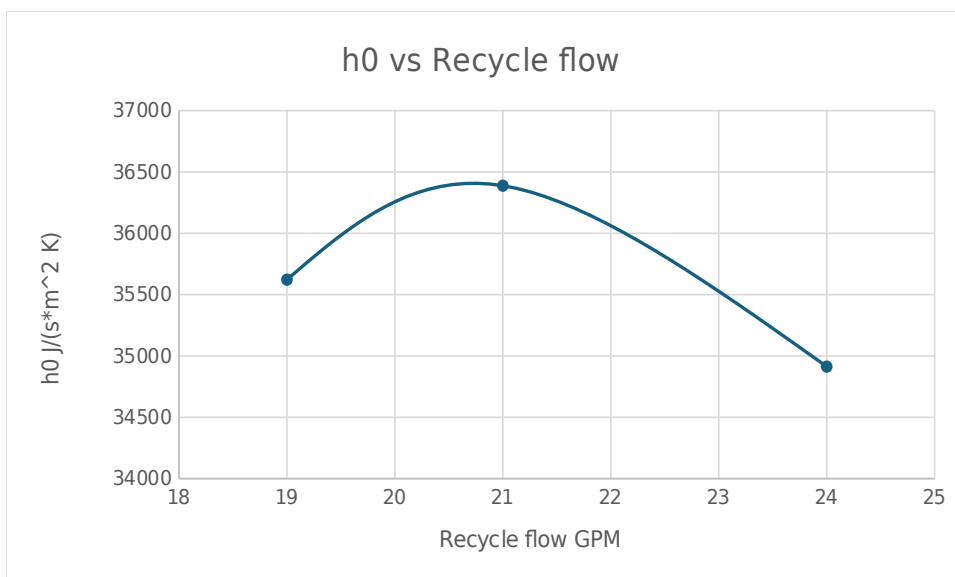
Figure 144: U_i Vs. Recirculation flow rate with bafflesFigure 25: h_o Vs Recirculation flow rate with baffles

Table 7 Unbaffled Trial 1 time, RPM, T1, and T2

Trial 1				
Time (sec)	Time, min	RP M	Inlet Coil Temp, T1, F	Tank Temp, T2, C
1.67E-02	0	310	44	50
8.33E-02	5	317	45	54
1.67E-01	10	323	46	56
2.50E-01	15	330	48	61
3.33E-01	SS	340	50	65

Table 8 Unbaffled Trial 1 T3 – T6

Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	T6, F
89	42	44	49
91	43	46	50
91	43	47	50
92	45	49	53
94	47	50	54

Table 9 Unbaffled Trial 1 T7, cooling flow rate, and recycle flow rate

outlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
47	18	58
47	21	59
48	22	60
50	24	60
52	25	62

Table 10 Unbaffled Trial 1 Flow rate of cooling water to recycle and time for collected steam

Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
0.5	
0.5	
0.5	
0.5	
0.5	36

Table 11 Unbaffled Trial 1 Volume of steam collected and the calculated steam flow rate

Steam - Volume, mL	Steam Flow Rate, L/sec
9.6	2.67E-04

Table 12 Unbaffled Trial 1 Qcoil, Qhx, Q13, and U

Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
1.35E+05	2.64E+02	1.35E+05	6.60E+03
1.63E+05	3.95E+02	1.63E+05	8.54E+03
1.64E+05	5.27E+02	1.65E+05	8.92E+03
1.72E+05	5.27E+02	1.73E+05	1.03E+04
1.76E+05	3.95E+02	1.76E+05	1.13E+04

Table 13 Unbaffled Trial 1 h_0 , $1/h_i$, h_i , viscosity, and Re impeller

h_0 , J/(s*m ² K)	$1/h_i$	h_i J/(s*m ² K)	viscosity, Pa*s	Re imp
300522	1.48E-04	6.75E+03	5.47E-04	9.74E+04
300522	1.14E-04	8.79E+03	5.12E-04	1.06E+05
300522	1.09E-04	9.20E+03	4.96E-04	1.12E+05
300522	9.42E-05	1.06E+04	4.59E-04	1.24E+05
300522	8.49E-05	1.18E+04	4.32E-04	1.35E+05

Table 14 Unbaffled Trial 1 k, Pr, NuT, and the calculated h_i from NuT

k, water (W/ m K)	Pr	NUT	h_i from nut, J/(s*m ² K)
6.40E-01	3.58E+00	1.17E+03	3.20E+03
6.30E-01	3.40E+00	1.22E+03	3.29E+03
6.30E-01	3.29E+00	1.24E+03	3.37E+03
6.50E-01	2.95E+00	1.28E+03	3.58E+03
6.60E-01	2.74E+00	1.33E+03	3.76E+03

Table 15 Unbaffled Trial 2 time, RPM, T1, and T2

	Time, min	RP M	Inlet Coil Temp, T1, F	Tank Temp, T2, C
1.67E-02	0	489	44	63
8.33E-02	5	500	47	61
1.67E-01	10	513	49	55
2.50E-01	15	517	49	56
3.33E-01	SS	520	51	57

Table 16 Unbaffled Trial 2 T3-T6

Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	T6, F
95	39	40	50
96	40	42	51.3
97	41	43.2	52
97	42	47	53
98	42.5	48	54

Table 17 Unbaffled Trial 2 T7, cooling flow rate, and recycle flow rate

Inlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
49	14	49
50	15	51.5
51.4	16.4	53

51.6	17.3	54
52	18	55

Table 18 Unbaffled Trial 2 Flow rate of cooling water to recycle and time for collected steam

Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
0.5	
0.5	
0.5	
0.5	
0.5	28.15

Table 19 Unbaffled Trial 2 Volume of steam collected and the calculated steam flow rate

Steam - Volume, mL	Steam Flow Rate, L/sec
6.5	2.31E-04

Table 20 Unbaffled Trial 2 Qcoil, Qhx, Q13, and U

Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
1.05E+05	1.32E+02	1.05E+05	6.47E+03
1.08E+05	2.64E+02	1.08E+05	6.41E+03
1.15E+05	2.90E+02	1.16E+05	6.16E+03
1.22E+05	6.59E+02	1.22E+05	6.63E+03
1.24E+05	7.25E+02	1.25E+05	6.87E+03

Table 21 Unbaffled Trial 2 h0, 1/hi, hi, viscosity, and Re impeller

h0, J/(s*m ² K)	1/hi	hi J/(s*m ² K)	viscosity, Pa*s	Re imp
3.15E+05	1.51E-04	6.60E+03	4.45E-04	1.89E+05
3.15E+05	1.53E-04	6.54E+03	4.59E-04	1.88E+05
3.15E+05	1.59E-04	6.28E+03	5.04E-04	1.75E+05
3.15E+05	1.48E-04	6.77E+03	4.96E-04	1.79E+05
3.15E+05	1.42E-04	7.03E+03	4.88E-04	1.83E+05

Table 22 Unbaffled Trial 2 k, Pr, NuT, and the calculated hi from NuT

k, water (W/ m K)	Pr	NUT	hi from nut J/(s*m ² K)
6.50E-01	2.86E+00	1.68E+03	4.70E+03

6.50E-01	2.95E+00	1.69E+03	4.72E+03
6.45E-01	3.27E+00	1.67E+03	4.63E+03
6.45E-01	3.22E+00	1.69E+03	4.68E+03
6.45E-01	3.17E+00	1.71E+03	4.72E+03

Table 23 Unbaffled Trial 3 time, RPM, T1, and T2

	Time, min	RP M	Inlet Coil Temp, T1, F	Tank Temp, T2, C
1.67E-02	0	720	52	56
8.33E-02	5	731	54	54
1.67E-01	10	738	55	53
2.50E-01	15	741	57	51
3.33E-01	SS	745	60	51

Table 24 Unbaffled Trial 3 T3 – T6

Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	Outlet Steam CW Temp, T6, F
76	82	69	48
78	84	71	50
79	85	73	52
80	88	74	54
82	90	75	55

Table 25 Unbaffled Trial 3 T7, cooling flow rate, and recycle flow rate

Inlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
46	0	58
48	0	60
50	0	61
52	0	63
52	0	63

Table 26 Unbaffled Trial 3 Flow rate of cooling water to recycle and time for collected steam

Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
0.5	
0.5	
0.5	
0.5	
0.5	24.93

Table 27 Unbaffled Trial 3 Volume of steam collected and the calculated steam flow rate

Steam - Volume, mL	Steam Flow Rate,
--------------------	------------------

	L/sec
5.2	2.09E-04

Table 28 Unbaffled Trial 3 Qcoil, Qhx, Q13, and U

Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
3.52E+04	-1.71E+03	3.35E+04	1.81E+03
4.23E+04	-1.71E+03	4.06E+04	2.12E+03
4.93E+04	-1.58E+03	4.77E+04	2.45E+03
5.40E+04	-1.85E+03	5.21E+04	2.59E+03
5.48E+04	-1.98E+03	5.29E+04	2.63E+03

Table 29 Unbaffled Trial 3 h0, 1/hi, hi, viscosity, and Re impeller

h0, J/(s*m ² K)	1/hi	hi J/(s*m ² K)	viscosity, Pa*s	Re imp
3.26E+05	5.49E-04	1.82E+03	4.96E-04	2.50E+05
3.26E+05	4.68E-04	2.14E+03	5.12E-04	2.46E+05
3.26E+05	4.05E-04	2.47E+03	5.21E-04	2.44E+05
3.26E+05	3.82E-04	2.61E+03	5.38E-04	2.37E+05
3.26E+05	3.77E-04	2.65E+03	5.38E-04	2.38E+05

Table 30 Unbaffled Trial 3 k, Pr, NuT, and the calculated hi from NuT

k, water (W/ m K)	Pr	NUT	hi from nut J/(s*m ² K)
6.47E-01	3.21E+00	2.11E+03	5.85E+03
6.43E-01	3.33E+00	2.11E+03	5.82E+03
6.40E-01	3.40E+00	2.11E+03	5.81E+03
6.40E-01	3.52E+00	2.10E+03	5.76E+03
6.40E-01	3.52E+00	2.10E+03	5.78E+03

Table 31 Unbaffled Trial 1, 2, and 3 Cooling flow rate and volumetric flow rate

	Cooling Coil Flow Rate, %GPM	volumetric flow m ³ /s
Trial 1		
	18	1.14E-03
	21	1.32E-03
	22	1.39E-03
	24	1.51E-03
	25	1.58E-03

Trial 2	Cooling Coil Flow Rate, %GPM	m ³ /s
	14	8.83E-04
	15	9.46E-04
	16.4	1.03E-03
	17.3	1.09E-03
	18	1.14E-03
Trial 3	Cooling Coil Flow Rate, %GPM	
	10	6.31E-04
	12	7.57E-04
	14	8.83E-04
	16	1.01E-03
	17	1.07E-03

Table 32 Unbaffled Trial 1, 2, and 3 mass flow rate of cooling coil

Trial 1	mass flow cooling coil kg/s
	1.14E+00
	1.32E+00
	1.39E+00
	1.51E+00
	1.58E+00
Trial 2	8.83E-01
	9.46E-01
	1.03E+00
	1.09E+00
	1.14E+00
Trial 3	6.31E-01
	7.57E-01
	8.83E-01
	1.01E+00
	1.07E+00

Table 33 Baffled Trial 1, 2, and 3 time, RPM, T1, and T2

Trial 1	Time, min	RPM	Inlet Coil Temp, T1, F	Tank Temp, T2, C
	0	337	62	42
	5	335	62.3	43.4
	10	338	64	44
	15	338.7	64.2	44.7
	SS	340	65	45
Trial 2	Time, min	RPM	Inlet Coil Temp, T1, F	Tank Temp, T2, C
	0	489	61.2	40.1
	5	493	62.3	40.3
	10	497.9	64	42
	15	500	64.2	43.3
	ss	500	65	44
Trial 3	Time, min	RPM	Inlet Coil Temp, T1, F	Tank Temp, T2, C
	0	690	61	40.2
	5	691	62	41
	10	691.3	63.5	42.4
	15	693	64.7	43
	ss	694	65	43.9

Table 34 Baffled Trials 1, 2, and 3 T3 – T6

Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	Outlet Steam CW Temp, T6, F
72	96	107.4	50.2
72.3	96.3	108	51.6
74	96.4	108.9	53.6
74.7	98.9	110.3	53.9
75	100	110	54
Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	Outlet Steam CW Temp, T6, F
70.2	96.4	107.4	50.2
70.3	97	109.4	51.3
72.4	98.3	109	52.6
73	99	110	53.5

75	100	110	54
Outlet Coil Temp, T3, F	Outlet Recycle Temp, T4, C	Inlet Recycle Temp, T5, C	Outlet Steam CW Temp, T6, F
59	96.3	107.3	49
59.2	96.6	107.6	49.2
61.3	96.5	108	51.4
61.4	97	109.6	51.9
62	97	110	52

Table 35 Baffled Trials 1,2, and 3 T7, cooling coil flow rate, and recycle flow rate

Inlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
46.3	0	
46.9	0	
47.8	0	
49	0	
50	0	19
Inlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
47.9	0	
48.2	0	
48.9	0	
49.6	0	
50	0	21
Inlet Steam CW Temp, T7, F	Cooling Coil Flow Rate, %GPM	Recycle Flow Rate, %GPM
47.2	0	
47.6	0	
48	0	
49	0	
50	0	24

Table 36 Baffled Trials 1, 2, and 3 Flow rate of cooling water to recycle and time for collected steam

Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
---	----------------------

0.5	
0.5	
0.5	
0.5	
0.5	15.8
Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
0.5	
0.5	
0.5	
0.5	
0.5	14.86
Cooling water to recycle heat exchanger Flow Rate, GPM	Steam - Time, sec
0.5	
0.5	
0.5	
0.5	
0.5	12.25

Table 37 Baffled Trials 1, 2, and 3 volume of steam and calculated steam flow rate

Steam - Volume, mL	Steam Flow Rate, L/sec
3.4	
	2.15E-04
Steam - Volume, mL	Steam Flow Rate, L/sec
3	2.02E-04

Steam - Volume, mL	Steam Flow Rate, L/sec
2.8	2.29E-04

Table 38 Baffled Trials 1, 2, and 3 Qcoil, Qhx, Q13, and U

Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
0.00E+00	1.50E+03	1502	65.
0.00E+00	1.54E+03	1542	68.
0.00E+00	1.65E+03	1647	73.
0.00E+00	1.50E+03	1502	67.
0.00E+00	1.32E+03	1317	59.
Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
0.00E+00	1.45E+03	1.45E+03	6.13E+01
0.00E+00	1.63E+03	1.63E+03	6.93E+01
0.00E+00	1.41E+03	1.41E+03	6.13E+01
0.00E+00	1.45E+03	1.45E+03	6.42E+01
0.00E+00	1.32E+03	1.32E+03	5.89E+01
Qcoil, J/s	QHX J/s	Q13 J/s	U (J/s*m ² *K)
0.00E+00	1.45E+03	1.45E+03	6.14E+01
0.00E+00	1.45E+03	1.45E+03	6.21E+01
0.00E+00	1.52E+03	1.52E+03	6.62E+01
0.00E+00	1.66E+03	1.66E+03	7.32E+01
0.00E+00	1.71E+03	1.71E+03	7.65E+01

Table 39 Baffled Trials 1, 2, and 3 h0, 1/hi, and hi

h0 J/(s*m ² K)	1/hi	hi J/(s*m ² K)
35622	0.015	65.

35622	0.014	68.
35622	0.013	73.
35622	0.014	67.
35622	0.016	59.
$h_0 \text{ J/(s}\cdot\text{m}^2 \text{ K)}$	$1/h_i$	$h_i \text{ J/(s}\cdot\text{m}^2 \text{ K)}$
$3.64\text{E}+04$	$1.63\text{E}-02$	$6.14\text{E}+01$
$3.64\text{E}+04$	$1.44\text{E}-02$	$6.95\text{E}+01$
$3.64\text{E}+04$	$1.63\text{E}-02$	$6.14\text{E}+01$
$3.64\text{E}+04$	$1.56\text{E}-02$	$6.43\text{E}+01$
$3.64\text{E}+04$	$1.69\text{E}-02$	$5.90\text{E}+01$
$h_0 \text{ J/(s}\cdot\text{m}^2 \text{ K)}$	$1/h_i$	$h_i \text{ J/(s}\cdot\text{m}^2 \text{ K)}$
$3.49\text{E}+04$	$1.63\text{E}-02$	$6.15\text{E}+01$
$3.49\text{E}+04$	$1.61\text{E}-02$	$6.22\text{E}+01$
$3.49\text{E}+04$	$1.51\text{E}-02$	$6.63\text{E}+01$
$3.49\text{E}+04$	$1.36\text{E}-02$	$7.33\text{E}+01$
$3.49\text{E}+04$	$1.30\text{E}-02$	$7.67\text{E}+01$

Table 40 Baffled Trials 1, 2, and 3 Cooling flow rate and volumetric flow rate

Trial 1	Cooling Coil Flow Rate, %GPM	m^3/s
	10	$6.31\text{E}-04$
	12	$7.57\text{E}-04$
	14	$8.83\text{E}-04$
	16	$1.01\text{E}-03$
	18	$1.14\text{E}-03$
Trial 2	Cooling Coil Flow Rate, %GPM	m^3/s

	9	5.68E-04
	10	6.31E-04
	13	8.20E-04
	14	8.83E-04
	15	9.46E-04
Trial 3	Cooling Coil Flow Rate, %GPM	
	10	6.31E-04
	12	7.57E-04
	15	9.46E-04
	16	1.01E-03
	16.5	1.04E-03

Table 41 Temperature losses between baffled and unbaffled trials

Temperature Loss between exp1 and exp2 (%)	
	0.16
	0.20
	0.21
	0.267
	0.308

Temperature Loss between exp1 and exp2 (%)	
	0.36
	0.34
	0.24
	0.23
	0.23

Temperature Loss between exp1 and exp2 (%)	
	0.28
	0.24
	0.20
	0.16
	0.14

8. Appendix II: Error Analysis (10 pts)

Quantitative

As mentioned in the discussion section, several constraints affected the results. The cooling water was extremely cold due to weather conditions and was therefore shut off, while clogged steam piping limited steam flow, preventing the tank from reaching the target temperature range and making steady-state conditions difficult to achieve. These constraints could've caused many points of errors when conducting heat transfer. One is the tank walls being steel, a very conductive metal, which becomes very cold and maintains a cold wall, minimizing heat transfer and causing a variance or steady values with an increase in steam flow.

Additionally, even beyond 1-1.5 turns of steam, the system struggled to raise and maintain the temperature, as it kept decreasing. The ideal temperature range of 65 °C – 85 °C was never reached, which could've caused issues with heat transfer.

Other possible errors can be the temperature gauges on the apparatus, with some being in Celsius and others in Fahrenheit. Because the scales differ, the temperatures can be slightly inaccurate when converted between Celsius and Fahrenheit.

Qualitative

EXP1			EXP2		
	Trial1 (340RPM) Tank Temp, T2, C		Trial 1 (340RPM) Tank Temp, T2, C		Temperature Loss between exp1 and exp2 (%)
	50		42		0.16
	54		43.4		0.20
	56		44		0.21
	61		44.7		0.267
	65		45		0.308
	Trial 2 (520 RPM) Tank Temp, T2, C		Trial 2 (500 RPM) Tank Temp, T2, C		Temperature Loss between exp1 and exp2 (%)
	63		40.1		0.36
	61		40.3		0.34
	55		42		0.24
	56		43.3		0.23
	57		44		0.23
	Trial 3 (745 RPM) Tank Temp, T2, C		Trial 3 (694 RPM) Tank Temp, T2, C		Temperature Loss between exp1 and exp2 (%)
	56		40.2		0.28
	54		41		0.24
	53		42.4		0.20
	51		43		0.16
	51		43.9		0.14

Figure 14. Temperature Loss/Drop between EXP1 and EXP2 (%)

Between experiment 1 and experiment 2, by trial, an observed drop of 20%-36% in temperature due to the system failing to maintain temperature even after shutting off the cooling water and cranking the steam past 1.5 turns.

9. Appendix III: Sample Calculations (10 pts)

The data set used for sample calculation is the unbaffled first trial with an RPM of 310.

[3] and [4] are used to obtain the thermal conductivity and viscosity of water.

To begin with, since some temperatures are in F rather than C, all temperatures are converted to C. An online converter was used.

Next, the steam flow rate is found by dividing volume by time:

Next, the mass flow of the cooling coil is found by dividing the volumetric flow rate by the specific volume of water.

The volumetric flow rate is:

The specific volume of water is:

Finally, the mass flow rate is:

From here, the can be found by using the following equation:

Plugging in the values from the run:

Cp of water is: 4184

Next, is found using the following formula:

is a known value of 0.0315

Plug in the values from this trial:

Next, is found by adding and :

Next, U is found by using the following equation:

Plug in the values from the trial:

Next, h_0 is found using the following equation:

Values of variables:

$k =$

$= 0.45 \text{ m}$

$L = 0.407 \text{ m}$

$= 0.000282$

Plugging in the values into the equation provides:

Next, μ is found using the following equation:

Plugging in the values provides:

All previous steps are equivalent for baffled and unbaffled runs. From here on, the steps shown only apply to unbaffled runs. Baffled runs are completed at this point.

For the unbaffled runs, the next steps include finding impeller Reynolds number, Prandtl fluid number, tank Nusselt number, and finally finding using the found numbers.

For formula for impeller Reynolds number is:

Where:

$= 0.1016 \text{ m}$

$n = 310 \text{ RPM}$

$=$

(at 50 C) =

Plugging in the values grants:

Next, the Prandtl fluid number is found using the following equation:

Where:

$C_p =$
(at 50 C) =
 k (at 50 C) =

Plugging in the values provides:

Next, the tank Nusselt number is found using the following equation:

Plugging in the previously found values for Reynolds and Prandtl number provides:

Lastly, can be calculated using the found numbers. The formula is:

Solving for :

Plugging in the values provides:

This concludes the sample calculations

10. Appendix IV: Individual Team Contributions

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
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Operator (Both Lab Days)	4	Attended lab
Pre-Lab Editing	1	Edited procedure
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	3	Worked on this part
Project Safety Evaluation (prelab)	0	
Final Lab Editing	2	Edited parts worked on
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	1	Added used references
Data Tabulation/Graphs (final)	3	Worked on data tabulation
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	3	Did calculations
Power Point Presentation	1	
Total	19	

NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended lab sessions
Pre-Lab Editing		
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	.5	Completed safety portion of the safety prelab.
Final Lab Editing	2	Edited theory and results section.
Executive Summary (final lab)	0	
Introduction (Final lab)	0	

Theory (Final Lab)	2	Explained how and what equations were used to calculate the numerical values.
Results (final lab)	4	Included data and graph figures explaining the trends in data.
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	3	Calculated values, tabulated data, and formed graphs
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	.5	Worked on safety and Challenges slide
Total	15	

NAME: Anna Cai

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Attended the lab period
Pre-Lab Editing	1	
Experimental Objective (Prelab)	.5	
Apparatus (prelab)	1	Wrote descriptions
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	1	Fixed grammar
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	1	Edited
Results (final lab)	3	Completed calculations
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	.5	
Data Tabulation/Graphs (final)	1	Formatted tables

Error Analysis (final lab)		
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Apparatus and materials
Total	16	

NAME: Hussein Rezk

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Operated the stirred tank
Pre-Lab Editing	1	Experimental
Experimental Objective (Prelab)	1	Experimental
Apparatus (prelab)		
Materials & Supplies (prelab)	1	Listed required material
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	5	Discussed results and correlations
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	Conducted qualitative and quantitative analysis
Sample Calculations (final Lab)		
Power Point Presentation	2	Theory sections
Total	16	